

SIAM AREFIN

Email | [GitHub](#) | [LinkedIn](#) | [Portfolio](#)

Research Interests: AI for Healthcare and Agriculture, Computer Vision, and Applied Machine Learning.

Academic Credentials

Shahjalal University of Science and Technology (SUST), Bangladesh
Bachelor of Science in Software Engineering

Mar 2022 – Jul 2026
CGPA: 3.10/4.00

Research Experiences

LLDDet: An Annotated Lemon Leaf Disease Dataset for Object Detection and Benchmarking with YOLOv11

Undergraduate Thesis — Supervisor: Fazle Rabbi Rakib

- Developed a deep learning-based object detection system for identifying and localizing diseases in lemon leaves using the LLDDet dataset.
- Achieved a test mAP of **0.927** while training under constrained computational resources with a batch size of 4 for 27 epochs.
- Investigated model generalization and object detection performance for real-world agricultural disease identification.

omicML: An Integrative Bioinformatics and Machine Learning Framework for Transcriptomic Biomarker Identification *Machine Learning Research Engineer* | Collaboration with Department of Biochemistry and Molecular Biology (BMB), SUST

- Implemented scalable machine learning deployment and monitoring pipelines using MLOps best practices.
- Collaborated with bioinformatics researchers to integrate computational modeling with biological large-data analysis.
- Worked on an interdisciplinary research problem connecting machine learning with transcriptomic biomarker identification.

Publications

- Siam Arefin, Md. Farzin Hossen, Fazle Rabbi Rakib, “LLDDet: An Annotated Lemon Leaf Disease Dataset for Object Detection and Benchmarking with YOLOv11,” [Manuscript in Preparation](#), 2026.
- Joy Prokash Debnath, Kabir Hossen, Md. Sayeam Khandaker, Shawon Majid, Md. Mehrajul Islam, **Siam Arefin**, Preonath Chondrow Dev, Saifuddin Sarker, Tanvir Hossain, “omicML: An Integrative Bioinformatics and Machine Learning Framework for Transcriptomic Biomarker Identification,” *bioRxiv*, 2025.

Preprint: [doi:10.1101/2025.10.25.684517](https://doi.org/10.1101/2025.10.25.684517)

Industry Experience

Kite Games Studio

Nov 2025 – Jan 2026

Software Engineer (ML Team) Intern | Mohakhali DOHS, Dhaka — Onsite

- Conducted R&D in image processing, machine learning, and generative AI for production-oriented applications.
- Worked with Diffusion and FLUX-based generative AI models for image generation and processing.
- Contributed to an Android application with 5M+ downloads and 700K+ daily active users.

Academic and Competitive Achievements

- **1st Place** – **Multi-Instance Object Detection Challenge**, [Duality AI International Competition](#); developed YOLO-based detection with ensemble learning and Sim2Real adaptation, achieving a **0.995 private leaderboard score**.
- **1st Runner-up** – **ML Olympiad**, [CO₂ Emissions Prediction Challenge](#); developed an ML pipeline involving pre-processing, feature engineering, model training, and optimization.
- **4th Place** – **Robi Datathon 3.0 Qualifier, Problem 1**, [Robi](#); developed a purchase prediction solution using historical customer transaction data and product-level information.
- **4th Place** – **DL Enigma 1.0**, [SUST CSE Carnival 2024](#); developed and optimized a YOLOv11m model for vehicle detection.
- **5th Place** – **Geospatial Object Detection Challenge**, [Duality AI & LunateAI](#); trained aerial object detection models using 100% synthetic data and applied Sim2Real adaptation for real-world rural building detection, ranking **5th among 54 teams**.
- **Top 10** – **DL Sprint 3.0**, [Bengali AI Math Olympiad](#); developed prompt-engineering-based solutions for Bengali mathematical problems using the Qwen model.
- **Top 10** – **Bhashamul: Bengali Regional IPA Transcription**, [UIU Computer Club](#), [UIU CSE & Bengali.AI](#); developed an NLP/deep learning solution for Bengali regional dialect text-to-IPA transcription, contributing to low-resource Bengali language technology.
- **800+ Problems Solved** on [Codeforces](#) and other competitive programming platforms, covering algorithms, data structures, and problem solving.

Projects

SM Blocker – Chrome Extension

[GitHub](#)

- Developed a Chrome extension to block distracting social media content and reduce unnecessary browsing during focused work sessions.
- Implemented real-time content filtering and browser-side automation to detect and restrict unwanted content, demonstrating practical JavaScript-based problem solving for everyday productivity.

Multi-Instance Object Detection

[GitHub](#)

- Merged synthetic datasets from multiple sources and applied ensemble learning to reduce the Sim2Real domain gap.
- Implemented custom post-processing using NMS and confidence filtering and fine-tuned YOLO models to improve generalization to real-world images.

<https://github.com/siamarefin/HealthMonitor>

HealthMonitor – AI-Powered Health Monitoring App

[GitHub](#)

- Developed and deployed a MERN-stack application with a custom-trained machine learning model for cardiovascular disease prediction.
- Integrated the Google Gemini API for intelligent advice and developed a RAG-based system for personalized insights using medical records and health-related data.
- Implemented secure authentication using JWT.

AI Processing API – AI/ML Backend System

[GitHub](#)

- Designed a modular AI backend using layered architecture, Provider/Adapter pattern, and Dependency Injection.
- Built REST APIs for audio transcription and medical-document OCR with validation, structured extraction, testing, and Dockerized deployment.

Technical Skills

Machine Learning:	Machine Learning, Deep Learning, NLP, CNNs, RNNs, Transformers, PyTorch, TensorFlow, Feature Engineering
Generative AI / LLMs:	LLMs, Prompt Engineering, Fine-tuning, RAG, Vector Databases, LangChain, AI Agents, Hugging Face, Diffusion Models
Computer Vision:	OpenCV, YOLO, Object Detection, Image Processing and Generation, Ensemble Learning
Programming:	Python, C, C++, Java, JavaScript
AI Systems / MLOps:	FastAPI, REST APIs, Docker, Git/GitHub, CI/CD, Azure, SQL, MongoDB
Software Engineering:	OOP, SOLID, Design Patterns, DSA, Software Architecture, Unit Testing